

Automation | Azure

21 June 2026 08:24

AIM:

While testing Azure Automation Runtime Environment 7.x on Hybrid Workers, I encountered an unexpected issue.

A parent runbook was able to call a local child script successfully in PowerShell 5.1, but the same implementation failed in Runtime Environment 7.x with:

"The term '.\init-test-variables.ps1' is not recognized..."

This investigation explores why the behavior differs between Runtime Environment 5.1 and 7.x, how child runbooks are executed, and what changes are required when migrating older automation patterns.

Environment Used:

Cloud: Azure

Runtime Environment:

- PowerShell 5.1
- PowerShell 7.4
- PowerShell 7.6

Job Execution:

- Hybrid Worker
- Cloud (ACI)

Visual Architecture Diagram:

PowerShell 5.1

Main Runbook

```
|
+--> .\child.ps1
|
+--> Same Process
+--> Same Context
```

PowerShell 7.x

Main Runbook

```
|
+--> Start-AzAutomationRunbook
|
+--> Separate Job
+--> Separate Context
+--> Separate Authentication
```

Lab & Analysis:

Step 1: Created an Automation Account over the **Central India** region:

Home > Automation Accounts

SecOpsDev Automation Account

List jobs executed by this Automation Account. Summarize properties for this Automation Account. Show me runbook status for this Automation Account.

Search

Switch to Old Experience Delete Move Explore in VS Code Give feedback Refresh

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Process Automation
- Configuration Management
- Shared Resources

Control your job execution environment, manage Packages easily and update the Runtime version of your runbooks using Runtime environment. [Learn more](#)

Azure Automation is revising the service and subscription limits starting 13 January 2025 to ensure fair share of cloud resources for all users. [Learn more](#)

Essentials

Resource group (move)	: CyberSecurityTraining	Subscription ID	:
Location	: Central India	Status	: Active
Subscription (move)	:	Last modified	: 3/12/2026, 16:42:24
Tags (edit)	: Add tags		

Step 2: Using the below script for the both the 5.1 and 7.4/7.6 runtime environment:

init-test-variables_5-1 :: Script:

```
# Create a hashtable named $CommonVars to store values that can be shared
# with the parent runbook or other scripts.
$CommonVars = @{}

# Sample key-value pair.
# Key : Message
# Value : "Hello from child runbook"
Message = "Hello from child runbook"
}

# Return the hashtable to the calling runbook.
# The parent runbook can access the value using:
# $ReturnedVars.Message
return $CommonVars
```

Home

init-test-variables_5-1 (SecOpsDev/init-test-variables_5-1)

Runbook

Search

Start View Edit Link to schedule

Overview

- Activity log
- Tags
- Diagnose and solve problems
- Resource visualizer
- Resources
- Runbook settings

Essentials

Resource group	: CyberSecurityTraining
Location	: Central India
Subscription	:
Subscription ID	:
Account	: SecOpsDev
Tags (edit)	: Add tags

View Published Source

```
1 # Create a hashtable named $CommonVars to store values that can be shared
2 # with the parent runbook or other scripts.
3 $CommonVars = @{}
4
5 # Sample key-value pair.
6 # Key : Message
7 # Value : "Hello from child runbook"
8 Message = "Hello from child runbook"
9 }
10
11 # Return the hashtable to the calling runbook.
12 # The parent runbook can access the value using:
13 # $ReturnedVars.Message
14 return $CommonVars
```

Output:

Home > SecOpsDev | Runbooks > init-test-variables_5-1 (SecOpsDev/init-test-variables_5-1)

init-test-variables_5-1 6/21/2026, 09:51

Job

Resume Stop Suspend Feedback Refresh Export output

Essentials

Id	: 8b081a41-c80b-4593-840a-624ba909394d	Created	: 6/21/2026, 09:51:22
Status	: Completed	Last Update	: 6/21/2026, 09:51:41
Ran on	: TestingHRW	Runbook	: init-test-variables_5-1
Ran As	: User	Source snapshot	: View source snapshot

Input Output Errors Warnings All Logs Exception

Name	Value
Message	Hello from child runbook

MainRunbook_5-1 :: Script:

```
# Display the current working directory to verify where the runbook/script is executing from.
Write-Output "PWD = $(Get-Location)"
# List all PowerShell scripts (*.ps1) in the current directory.
# Useful for confirming that the child script is present and accessible.
Get-ChildItem . -Filter *.ps1 | Select Name
# Log a message indicating that the child script execution test is starting.
Write-Output "Testing local script call"
# Execute the child script and capture its output in the $ChildOutput variable.
$ChildOutput = .\init-test-variables_5-1.ps1
# Display the output returned by the child script.
Write-Output $ChildOutput
```

The screenshot shows the 'View Published Source' interface in the Azure portal. On the left, there's a sidebar with navigation options like 'Overview', 'Activity log', 'Tags', etc. The main area displays the PowerShell script for 'MainRunbook_5-1' with line numbers 1 through 15. The script includes comments and commands for displaying the current directory, listing PowerShell scripts, executing a child script, and displaying its output.

Output:

The screenshot shows the 'Output' tab for the runbook 'MainRunbook_5-1'. It displays the execution results, including the current directory path and the output of the child script. The output is as follows:

```
PWD = C:\ProgramData\Microsoft\System Center\Orchestrator\7.2\SMA\Sandboxes\vcdo4ad.t11\Temp\z05aqmb4.k1c

Name
----
init-test-variables_5-1.ps1

Testing local script call

Name      Value
----      -
Message   Hello from child runbook
```

Now executing the same runbook over the 7.4 runtime environment

init-test-variables :: Script:

MainRunbook 6/21/2026, 10:21

Job

[Resume](#)
[Stop](#)
[Suspend](#)
[Feedback](#)
[Refresh](#)
[Export output](#)

Essentials

Id	: e6063a39-5e26-4700-aa82-0389cff3ac0f	Created	: 6/21/2026, 10:21:10
Status	: Completed	Last Update	: 6/21/2026, 10:21:43
Ran on	: TestingHRW	Runbook	: MainRunbook
Ran As	: User	Source snapshot	: View source snapshot

[Input](#)
[Output](#)
[Errors](#)
[Warnings](#)
[All Logs](#)
[Exception](#)

Errors

7

Search logs...

Time	Type	Details
6/21/2026, 10:21:43	Error	
6/21/2026, 10:21:43	Error	
6/21/2026, 10:21:43	Error	
6/21/2026, 10:21:43	Error	
6/21/2026, 10:21:43	Error	
6/21/2026, 10:21:43	Error	

Job Stream:

```

6/21/2026, 10:21:42 - Output: PWD = C:\Packages\Plugins
\Microsoft.Azure.Automation.HybridWorker.HybridWorkerForWindows\1.3.76
\HybridWorkerPackage\HybridWorkerAgent
6/21/2026, 10:21:42 - Output: Testing local script call
6/21/2026, 10:21:43 - Error: [31;1m.\init-test-variables.ps1: [0mC:\ProgramData\Microsoft
\System Center\Orchestrator\7.2\SMA\Sandboxes\wdbbh3fmv.gqm\Temp\jdpqxbt3.ygr
\afa8041b-ada9-4b50-809c-dcd0432f6a94.ps1:9 [0m
6/21/2026, 10:21:43 - Error: [31;1m [0m [36;1mLine | [0m
6/21/2026, 10:21:43 - Error: [31;1m [0m [36;1m [36;1m 9 | [0m $ChildOutput =
[36;1m.\init-test-variables.ps1 [0m
6/21/2026, 10:21:43 - Error: [31;1m [0m [36;1m [36;1m [0m [36;1m [0m [36;1m [0m [36;1m |
[31;1m ~~~~~ [0m
6/21/2026, 10:21:43 - Error: [31;1m [0m [36;1m [36;1m [0m [36;1m [0m [36;1m [31;1m [31;1m [36;1m |
[31;1mThe term '.\init-test-variables.ps1' is not recognized as a name of a cmdlet, function,
script file, or [0m
6/21/2026, 10:21:43 - Error: [31;1m [0m [36;1m [36;1m [0m [36;1m [0m [36;1m [31;1m [31;1m [36;1m [31;1m
[36;1m | [31;1mexecutable program. Check the spelling of the name, or if a path was
included, verify that the path is correct [0m
6/21/2026, 10:21:43 - Error: [31;1m [0m [36;1m [36;1m [0m [36;1m [0m [36;1m [31;1m [31;1m [36;1m [31;1m
[36;1m [31;1m [36;1m | [31;1mand try again. [0m

```

Explanation:

Before the error occurred, the runbook printed the current working directory:

```

PWD = C:\Packages\Plugins
\Microsoft.Azure.Automation.HybridWorker.HybridWorkerForWindows\1.3.76
\HybridWorkerPackage\HybridWorkerAgent

```

The script then attempted to execute:

```
.\init-test-variables.ps1
```

When PowerShell encounters a relative path beginning with .\, it searches for the file in the current working directory. Since the working directory was the Hybrid Worker Agent folder and the file init-test-variables.ps1 did not exist in that location, PowerShell was unable to

find the script and threw the error.

To fix this there are 2 different ways:

- 2. This approach was used only for lab validation and is not recommended for production environments because it introduces a dependency on local files existing on the Hybrid Worker. (C:\Packages\Plugins\Microsoft.Azure.Automation.HybridWorker.HybridWorkerForWindows\1.3.76\HybridWorkerPackage\HybridWorkerAgent)**

Script:

```
# Create a hashtable named $CommonVars to store values that can be shared
# with the parent runbook or other scripts.
$CommonVars = @{

    # Sample key-value pair.
    # Key : Message
    # Value : "Hello from child runbook"
    Message = "Hello from child runbook"
}
# Return the hashtable to the calling runbook.
# The parent runbook can access the value using:
# $ReturnedVars.Message
return $CommonVars.Message
```

Output:

Home > MainRunbook (SecOpsDev/MainRunbook)



MainRunbook 6/21/2026, 10:37 ...

Job

▶ Resume □ Stop || Suspend 🗨 Feedback ↻ Refresh ⬆ Export output

^ Essentials

Id	: f3230fa0-1b78-4af4-a374-0797efb767d0	Created	: 6/21/2026, 10:37:41
Status	: Completed	Last Update	: 6/21/2026, 10:38:13
Ran on	: TestingHRW	Runbook	: MainRunbook
Ran As	: User	Source snapshot	: View source snapshot

Input Output Errors Warnings All Logs Exception

```
PSD = C:\Packages\Plugins\Microsoft.Azure.Automation.HybridWorker.HybridWorkerForWindows\1.3.76\HybridWorkerPackage\HybridWorkerAgent

B[32;1mNameB[0m
B[32;1m---B[0m
init-test-variables.ps1
Testing local script call
Hello from child runbook
```

1. Executing the child runbook as Azure Automation runbook:

As per Microsoft documentation <https://learn.microsoft.com/en-us/azure/automation/automation-child-runbooks#runbook-types>; only PowerShell 5.1 is supported for .\child-runbook.ps1.

Runbook types

Currently, PowerShell 5.1 is supported and only certain runbook types can call each other:

- A **PowerShell runbook** and a **graphical runbook** can call each other inline, because both are PowerShell based.
- A **PowerShell Workflow runbook** can call another PowerShell Workflow runbook inline.
- The PowerShell types and the PowerShell Workflow types can't call each other inline. They must use `Start-AzAutomationRunbook`.

❗ Important

- Executing child scripts using `.\child-runbook.ps1` is not supported in PowerShell 7.1 and PowerShell 7.2. **Workaround:** Use `Start-AutomationRunbook` ([internal cmdlet](#)) or `Start-AzAutomationRunbook` (from [Az.Automation module](#)) to start another runbook from parent runbook.
- Executing child scripts using Graphical PowerShell Workflow runbooks is not supported. **Workaround:** Use `Start-AzAutomationRunbook` (from [Az.Automation module](#)) from within the parent runbook (via an `InlineScript` block) to start another runbook.

Hence, updated the script as per the documentation: <https://learn.microsoft.com/en-us/powershell/module/az.automation/start-azautomationrunbook?view=azps-16.0.0> & <https://learn.microsoft.com/en-us/powershell/module/az.automation/get-azautomationjoboutput?view=azps-16.0.0>

Script:

```
# Display the current working directory to verify where the runbook/script is executing from.
Write-Output "PWD = $(Get-Location)"
# List all PowerShell scripts (*.ps1) in the current directory.
# Useful for confirming that the child script is present and accessible.
Get-ChildItem . -Filter *.ps1 | Select Name
# Log a message indicating that the child script execution test is starting.
#//Write-Output "Testing local script call"
# Execute the child script and capture its output in the $ChildOutput variable.
#//$ChildOutput = .\init-test-variables.ps1
# Display the output returned by the child script.
#//Write-Output $ChildOutput
# Connect to Azure using Managed Identity
Connect-AzAccount -Identity
# Variables
$ResourceGroupName = "CyberSecurityTraining"
$AutomationAccountName = "SecOpsDev"
$ChildRunbookName = "init-test-variables"
# Start the child runbook
Write-Output "Starting child runbook: $ChildRunbookName"
$Job = Start-AzAutomationRunbook `
    -ResourceGroupName $ResourceGroupName `
    -AutomationAccountName $AutomationAccountName `
    -Name $ChildRunbookName
Write-Output "Child runbook started successfully."
Write-Output "Job ID: $($Job.JobId)"
# Wait for completion
do {
    Start-Sleep -Seconds 5
    $JobStatus = Get-AzAutomationJob `
        -ResourceGroupName $ResourceGroupName `
        -AutomationAccountName $AutomationAccountName `
        -Id $Job.JobId
    Write-Output "Current Status: $($JobStatus.Status)"
} while ($JobStatus.Status -eq "Running" -or $JobStatus.Status -eq "New" -
or $JobStatus.Status -eq "Activating")
# Display final status
```

```

Write-Output "Final Status: $($JobStatus.Status)"
# Retrieve output records
$OutputRecords = Get-AzAutomationJobOutput `
  -ResourceGroupName $ResourceGroupName `
  -AutomationAccountName $AutomationAccountName `
  -Id $Job.JobId
# Retrieve actual output messages
Write-Output "==== CHILD RUNBOOK OUTPUT ====="
foreach ($Record in $OutputRecords)
{
  $Output = Get-AzAutomationJobOutputRecord `
    -ResourceGroupName $ResourceGroupName `
    -AutomationAccountName $AutomationAccountName `
    -JobId $Job.JobId `
    -Id $Record.StreamRecordId

  Write-Output $Output.Value
  Write-Output $Output.Value.Message
}

```

Output:

Home > SecOpsDev | Runbooks > MainRunbook (SecOpsDev/MainRunbook)


MainRunbook 6/21/2026, 22:01

Job

▶ Resume □ Stop || Suspend 🗨 Feedback ↻ Refresh 📄 Export output

Status	: Completed	Last Update	: 6/21/2026, 22:02:11
Ran on	: TestingHRW	Runbook	: MainRunbook
Ran As	: User	Source snapshot	: View source snapshot

Input **Output** Errors Warnings All Logs Exception

```

PS> C:\Packages\Plugins\Microsoft.Azure.Automation.HybridWorker.HybridWorkerForWindows\1.3.76\HybridWorkerPackage\HybridWorkerAgent

R[32;1mName][0m
R[32;1m----R[0m
init-test-variables.ps1

Starting child runbook: init-test-variables
Child runbook started successfully.
Job ID: b03b2969-297a-439f-b56c-77c9bc1434a8
Current Status: New
Current Status: New
Current Status: New
Current Status: Activating
Current Status: Activating
Current Status: Activating
Current Status: Activating
Current Status: Completed
Final Status: Completed
==== CHILD RUNBOOK OUTPUT =====
Message
Hello from child runbook

```

Observations and Key Findings

Observation 1: Parent and Child Runbooks Execute as Separate Jobs

When a parent runbook invokes a child runbook using Start-AzAutomationRunbook, Azure Automation creates a separate job for the child runbook.

This means:

- The parent runbook continues as one job.
- The child runbook executes as a separate job with its own Job ID.
- Output and status must be retrieved using the child job's Job ID.

Conceptually, the execution flow looks like:

Parent Runbook Job

|

```
+--> Start-AzAutomationRunbook
|
+--> Child Runbook Job
```

Therefore, this approach differs from a local PowerShell script call where execution remains within the same process and execution context.

Observation 2: Child Runbook Does Not Automatically Run on the Hybrid Worker

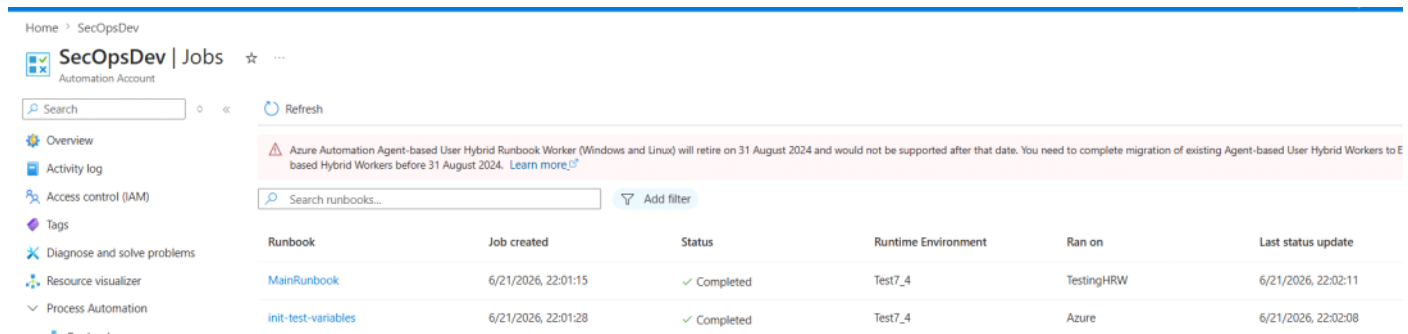
This behavior is especially important for runbooks that rely on local resources, installed software, network access, or credentials available only on a Hybrid Worker.

A significant finding during testing was that starting a child runbook from a parent runbook running on a Hybrid Worker does not automatically cause the child runbook to execute on the same Hybrid Worker.

Example:

```
Start-AzAutomationRunbook -Name "init-test-variables"
```

Without specifying a Hybrid Worker, Azure Automation determines the execution environment for the child runbook.



Runbook	Job created	Status	Runtime Environment	Ran on	Last status update
MainRunbook	6/21/2026, 22:01:15	✓ Completed	Test7_4	TestingHRW	6/21/2026, 22:02:11
init-test-variables	6/21/2026, 22:01:28	✓ Completed	Test7_4	Azure	6/21/2026, 22:02:08

To explicitly execute the child runbook on a Hybrid Worker, the **-RunOn** parameter should be provided:

```
Start-AzAutomationRunbook `
-Name "init-test-variables" `
-RunOn "HybridWorkerName"
```

Key takeaway:

Parent execution on a Hybrid Worker does not automatically imply child execution on the same Hybrid Worker. The execution location should be explicitly validated and configured when required.

Observation 3: Authentication Is Required Before Starting Child Runbooks

Testing confirmed that Start-AzAutomationRunbook requires an authenticated Azure context.

Without authentication:

```
Connect-AzAccount -Identity
```

the cmdlet is unable to communicate with Azure Automation and start the child runbook.

Typical pattern:

```
Connect-AzAccount -Identity
Start-AzAutomationRunbook ...
```

Key takeaway:

Starting another runbook is an Azure management operation and therefore requires a valid Azure authentication context.

Observation 4: Runtime Environment 7.x Breaks Traditional Shared Initialization Patterns

An interesting observation during testing was the difference in behavior between PowerShell 5.1 and PowerShell 7.x Runtime Environments.

Historically, many implementations relied on a shared initialization runbook that:

- Retrieved Automation Variables.
- Retrieved Key Vault secrets.
- Established Azure authentication.
- Returned commonly used values to other runbooks.

However, with Runtime Environment 7.x:

- Runbooks execute in more isolated execution contexts.
- Authentication established in one runbook is not automatically inherited by another runbook started through Start-AzAutomationRunbook.
- Each runbook should be treated as an independent execution unit.

Practical implication:

A child runbook that performs Connect-AzAccount does not automatically provide that authenticated session to the parent runbook or to other independently started runbooks.

Key Points

1. Start-AzAutomationRunbook creates a separate Azure Automation job.
2. Child runbooks have their own Job IDs, status, and output streams.
3. Parent and child runbooks should be treated as independent execution contexts.
4. Running a parent runbook on a Hybrid Worker does not guarantee the child will run on the same Hybrid Worker.
5. Use the -RunOn parameter when Hybrid Worker execution is required.
6. An authenticated Azure context is required before invoking Azure Automation management cmdlets. In Azure Automation this is commonly achieved using Managed Identity through Connect-AzAccount -Identity.
7. Authentication contexts are not automatically shared between independently executed runbooks.
8. Runtime Environment 7.x introduces stricter execution isolation compared to patterns commonly used in PowerShell 5.1 environments.

Why This Matters

Organizations often maintain a shared initialization runbook that:

- Retrieves Automation Variables
- Retrieves Key Vault secrets
- Performs Azure authentication
- Returns common configuration values

These patterns worked reliably in many PowerShell 5.1 implementations.

When migrating to Runtime Environment 7.x, engineers should validate these assumptions because child runbooks execute independently and do not automatically share authentication or execution context.

Lessons Learned

- Do not assume child runbooks inherit the parent's execution environment.
- Always verify where a child runbook executes.
- Treat each runbook as an independent automation workload.
- Validate PowerShell 5.1 patterns before migrating to Runtime Environment 7.x.
- Explicit authentication and execution targeting reduce troubleshooting complexity.

Final Conclusion

The issue was not a defect in Azure Automation Runtime Environment 7.x.

The failure occurred because the child script was not available within the Runtime Environment 7.x execution context. Traditional relative-path child script patterns that were commonly used in PowerShell 5.1 should be revalidated when migrating to Runtime Environment 7.x.

Instead:

- Child runbooks should be executed using Start-AzAutomationRunbook.
- Each child runbook must be treated as an independent job.
- Authentication should be established independently.
- Hybrid Worker execution should be explicitly defined using the -RunOn parameter when required.

Understanding these execution boundaries is critical when modernizing legacy PowerShell 5.1 automation solutions.